

Section 3 – Addressing the Ethical Issues of LLMs (part 2)

COMM116 Generative AI Applications

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Aspects of Ethical AI

- Fairness
 - Bias
 - Fairness
- Accountability
 - Physical safety
 - Data safety
 - Economic Safety
- Transparency
 - Explainable AI



A farsighted conference since 2018

Screenshot from <https://facctconference.org/index.html>

Fundamental issues of LLMs

- Hallucination
 - We will look at this on this Friday 13 March 2026.
- Opaqueness
- Staleness
- Incompleteness

(The above from Hogan et al., 2025)

- Ethical concerns: harmful, trustworthiness, uncertainty, etc.

Hogan, A., Dong, X. L., Vrandečić, D., & Weikum, G. (2025). Large Language Models, Knowledge Graphs and Search Engines: A Crossroads for Answering Users' Questions. *arXiv preprint arXiv:2501.06699*.

Pre-class targeted reading task (for 10th March)

Investigate one of the techniques or directions as assigned, read something about it:

1. Attention-based explainability (Sect. 3.1.2)
 - <https://aclanthology.org/P19-3007/>
2. Concept-based explainability (Sect. 3.2.3)
 - <https://aclanthology.org/2022.acl-long.581/> (& for Neuron Activation)
 - <https://doi.ieeecomputersociety.org/10.1109/CVPR52733.2024.01043>
3. ERASER, Faithfulness evaluation of explainability
 - <https://aclanthology.org/2020.acl-main.408/>
4. SHAP (Surrogate Models in Sect. 3.1.1)
 - <https://christophm.github.io/interpretable-ml-book/shap.html>
 - <https://aclanthology.org/2021.hackashop-1.3.pdf>
5. UMAP
 - <https://arxiv.org/abs/1802.03426>
 - <https://pair-code.github.io/understanding-umap/>
 - <https://doi.org/10.1038/s43586-024-00363-x>
6. Other directions: Probing-based (Sect. 3.2.1), Neuron activation (Sect. 3.2.2), mechanistic interpretability (Sect. 3.2.4), Natural Language Explanation (Sect. 3.1.4 and 3.2.2), etc.

Think about five aspects:

- Intuition of the method
- A use case
- An advantage
- A disadvantage
- Feasibility to implement

Section numbers are from the survey:
Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. Explainability for Large Language Models: A Survey. ACM Trans Intell Syst Technol. 2024 Feb 22;15(2):20:1-20:38.
<https://doi.org/10.1145/3639372>

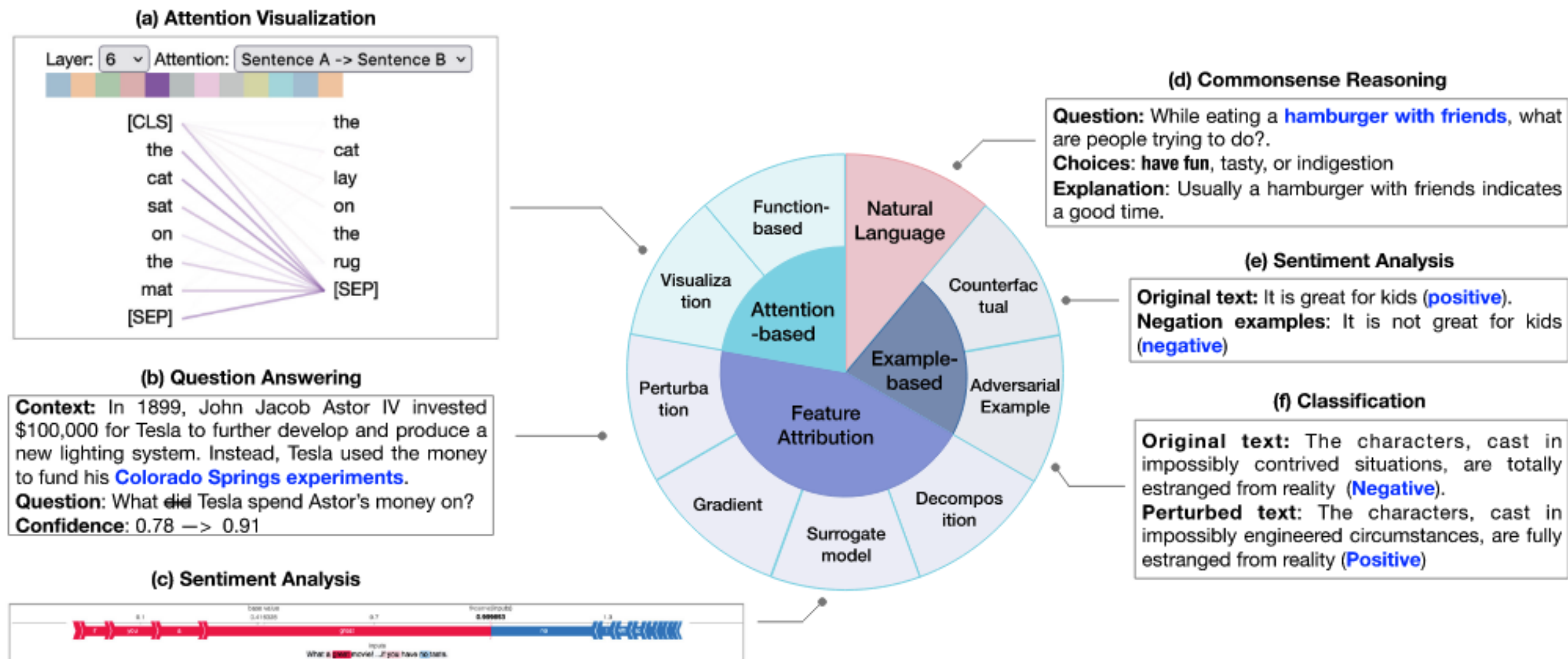


Fig. 3. Local explanation is composed of four subareas. The organization of each subarea and examples for certain individual explanation methodology have been given. (a) Bipartite graph attention representation

Explainability for LLMs

Local explanations

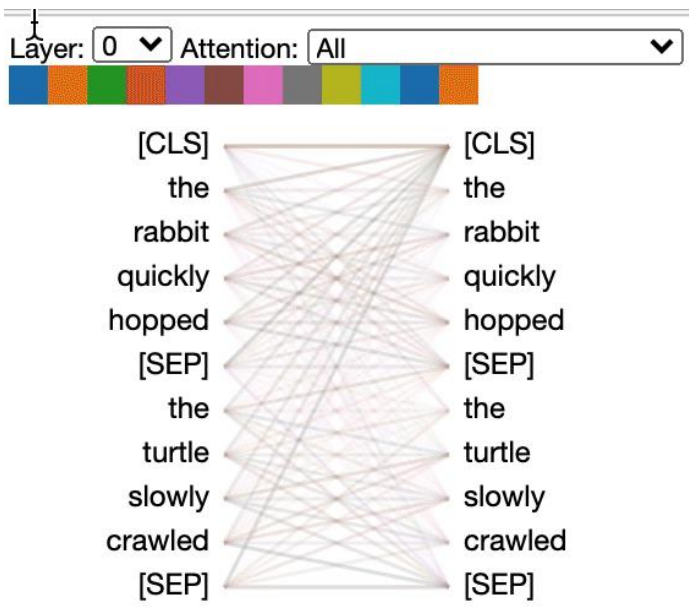
Feature Attribution: perturbation-based, gradient-based, **surrogate models (e.g. SHAP)**, decomposition-based

Attention-based explanation: visualization, function-based

Example-based: Adversarial examples, counterfactual explanations, data influence

Natural language explanation

Attention-based explanation



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Scaled Dot-Product Attention

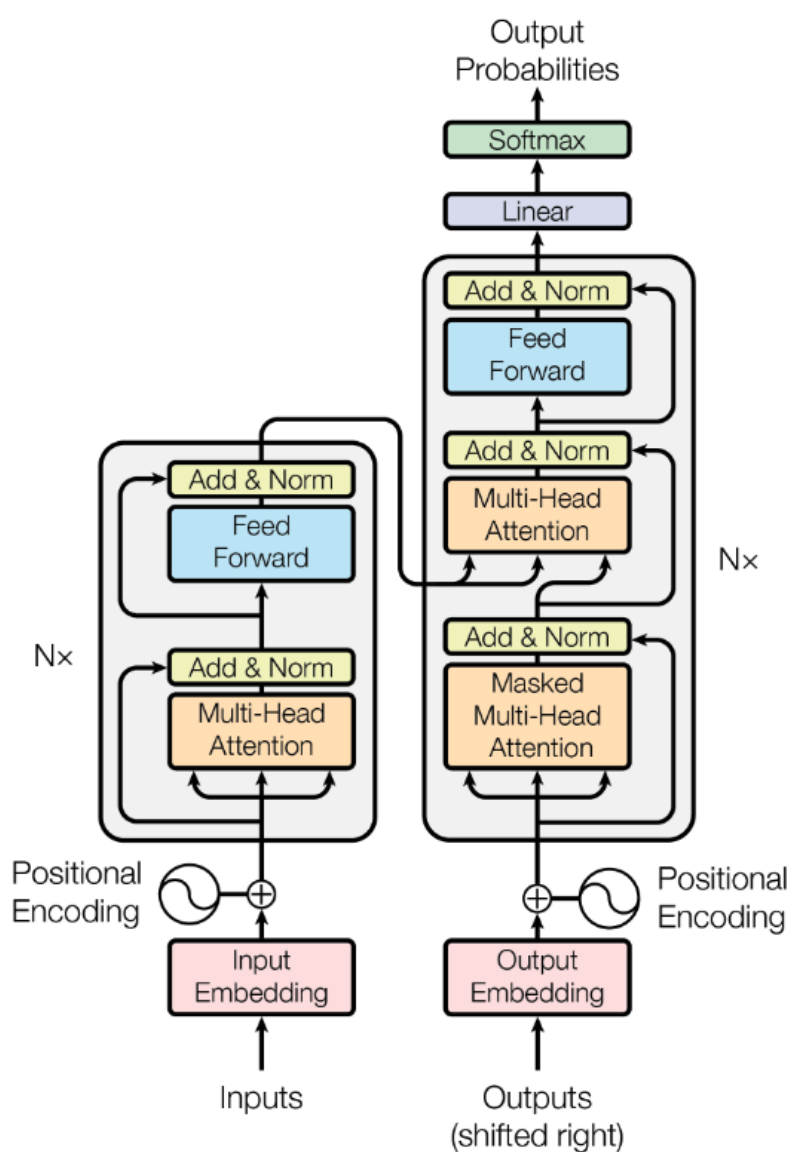
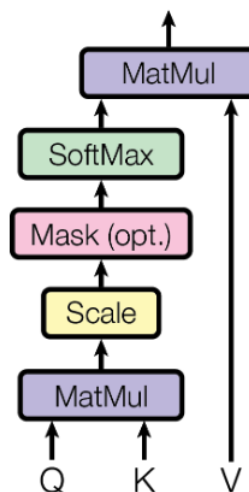


Figure 1: The Transformer - model architecture.

Image above from
<https://github.com/jessevig/bertviz?tab=readme-ov-file> and see ref below

Jesse Vig. 2019. [A Multiscale Visualization of Attention in the Transformer Model](https://arxiv.org/abs/1906.01615).
In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 37–42, Florence, Italy. Association for Computational Linguistics.

Images and equation from Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, 30.

Self-attention in BERT

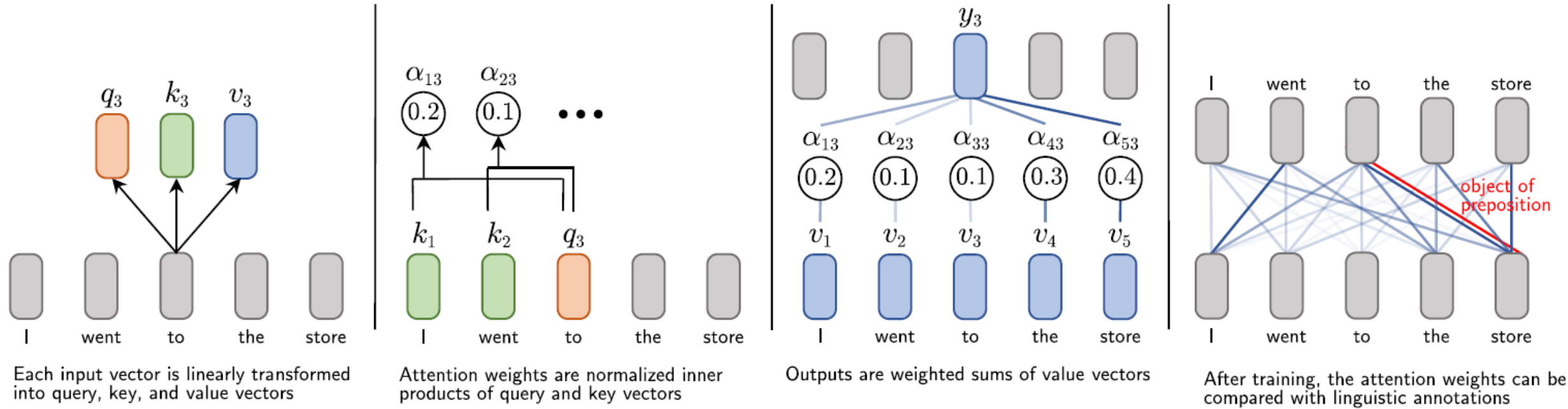


Fig. 5. An overview of the neural attention mechanism. NLP models often contain many attention mechanisms, each producing a different set of attention weights.

Attention heads as simple classifiers

$$\text{precision}(h) = \frac{1}{N} \sum_{w \in \text{corpus}} \sum_{j \in S_l(w)} l(w_{\text{argmax}_i \alpha(w, h)_i^j}, w_j), \quad [3]$$

tations assign each pair of words (w_i, w_j) a label $l(w_i, w_j)$ that is 1 if a particular linguistic relationship between words holds (e.g., w_i is w_j 's syntactic head) and is 0 if otherwise. To inter-

$\alpha(w, h)$ denotes the attention distribution of head h when BERT is run over the sequence of words $w = [w_1, \dots, w_n]$, we find the most-attended-to word $w_{\text{argmax}_i \alpha(w, h)_i^j}$ for each position $1 \leq j \leq n$.

$S_l(w) = \{j : \sum_{i=1}^n l(w_i, w_j) > 0\}$ is the subset of the input expressing the annotated relationship

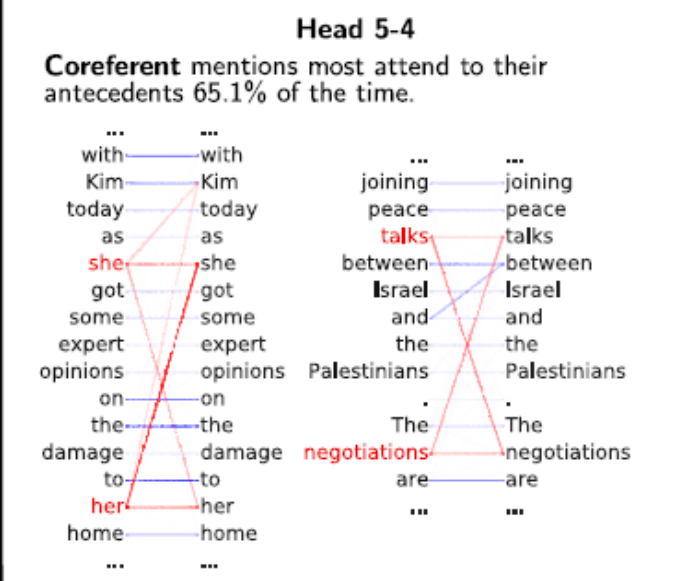
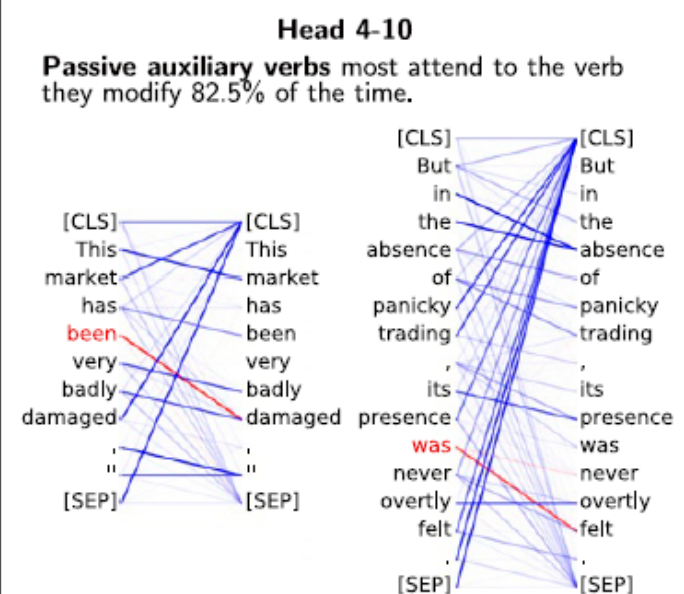
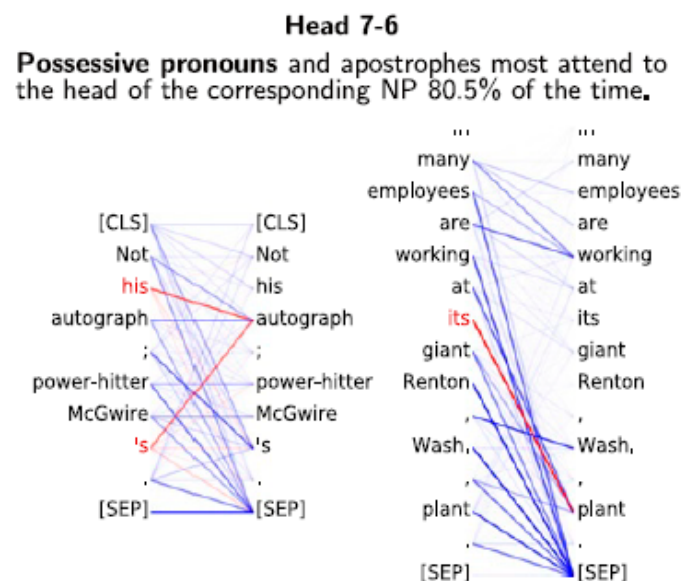
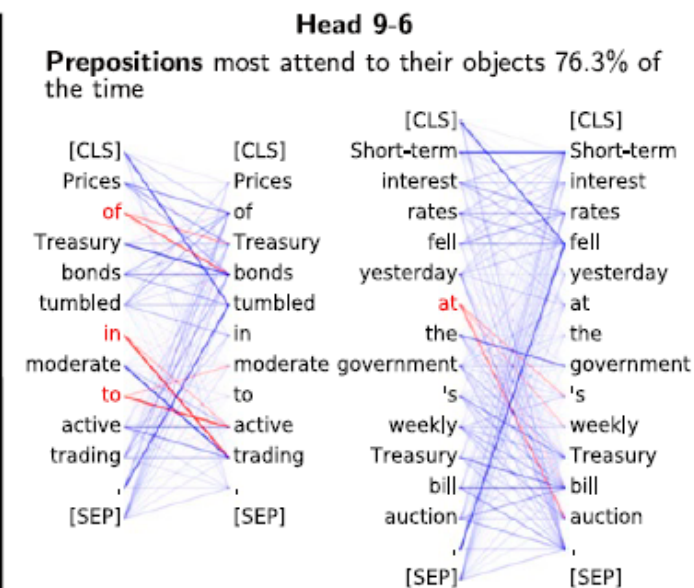
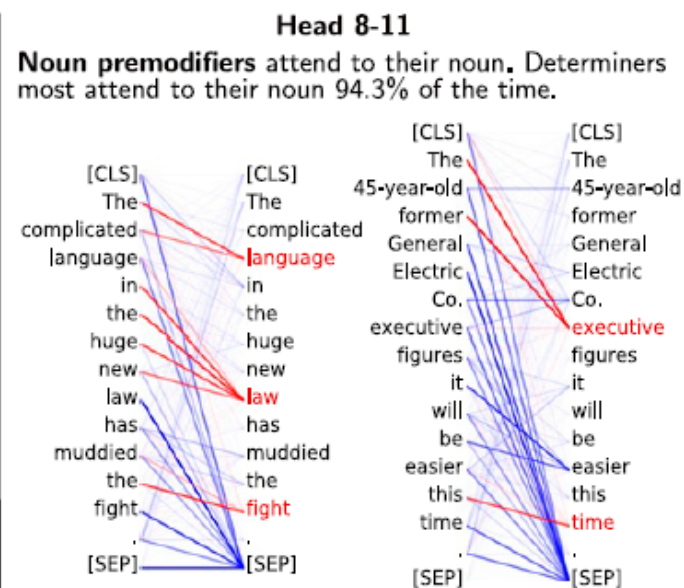
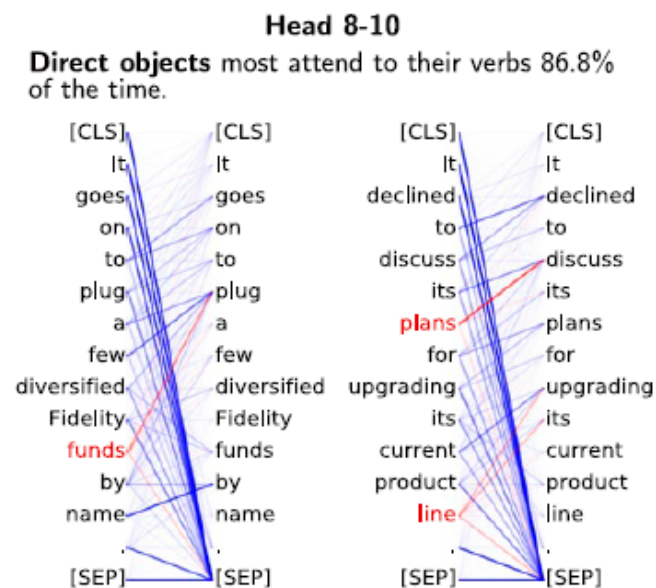


Figure from Manning et al., 2020 as the previous slide

Fig. 6. Some BERT attention heads that appear sensitive to linguistic phenomena, despite not being explicitly trained on linguistic annotations. In the example attention maps, the darkness of a line indicates the size of the attention weight. All attention to/from red words is colored red; these words are chosen to highlight certain of the attention heads' behaviors. [CLS] (classification) and [SEP] (separator) are special tokens BERT adds to the input during preprocessing. Attention heads are numbered by their layer and index in BERT. Reprinted with permission from ref. 59, which is licensed under [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/).

Attention Visualisation in Hierarchical Label-wise Attention Networks

For 427.31 For 428.0 Document #24 in MIMIC-III-50

0.02	0.01	admission	date	discharge	date	service	surgery						
0.01	0.01	allergies	patient	recorded	as	having	no	known	allergies	to	drugs	attending	
0.01	0.02	major	surgical	or	invasive	procedure	ex	lap	r	hemicolectomy	mucous	fistula	
0	0	history	of	present	illness	age	over	f	presented	to	location	un	
0	0.01	admitted	to	hospital1	and	taken	directly	to	or	upon	arrival		
0.41	0.8	past	medical	history	pmhx	a	fib	aortic	stenosis	chf	last	ef	
0	0	pertinent	results	00pm	blood	wbc	rbc	hgb	hct	mcv	mch	mchc	
0	0.01	brief	hospital	course	age	over	f	transferred	from	location	un	and	
0	0.02	patient	was	taken	directly	to	the	operating	room	for	an	exploratory	
0	0	she	underwent	a	right	hemicolectomy	mucous	fistula	ileostomy	gj	tube	placement	
0.01	0.01	fluid	balance	intraoperatively	included	units	ffp	units	plt	u	nits	prbc	
0	0	patient	was	kept	intubated	and	taken	directly	to	the	surgical	intensive	
0	0.02	she	required	maximum	pressor	support	to	maintain	sufficient	cardiac	index		
0	0.03	patient	did	show	signs	of	distal	ischemia	to	extremities	by	the	
0.01	0	family	meeting	at	latter	evening	decided	to	make	patient	cmo	patient	
0.01	0	medications	on	admission	last	name	un	amlodipine	mg	qd	benarepril	mg	
0.49	0	discharge	diagnosis	cardiopulmonary	arrest	perforated	colon	atrial	fibrillation	ventilatory	support	discharge	

Sentence-level attention weights

427.31 (Atrial Fibrillation)

428.0 (Congestive heart failure, unspecified)

Visualisation of attention weights in Hierarchical Label-wise Attention Networks (HLAN)

Image from Dong H, Suárez-Paniagua V, Whiteley W, Wu H. Explainable Automated Coding of Clinical Notes using Hierarchical Label-wise Attention Networks and Label Embedding Initialisation. *Journal of Biomedical Informatics*. 2021 Mar 9;103728. Preprint on arXiv: <https://arxiv.org/abs/2010.15728>

Debates in attention-based explanation

(as surveyed by Zhao et al., 2024)

Pros:

- Visualizing attentions provides an intuitive way to understand how models work by showing attention patterns and statistics.
- Function-based attention (e.g. gradient-based) works better
- Attention heads in BERT have been shown to encode syntax effectively

Cons:

- Attention often does not identify the most important features for prediction
- Inferior explanations compared to these alternatives
- Cannot be correlated with other explanation methods
- Limited usefulness of the attention mechanism in model predictions

Neuron activation explanation

- Neuron analysis looks into individual dimensions, i.e., neurons in representations, that are crucial for model performance or associated with specific linguistic properties. (Zhao et al., 2024)
- We use Knowledge Neuron as an example.
- You can also see Grad-CAM (<https://arxiv.org/pdf/1610.02391>) as another example for imaging.

Figure from Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. [Knowledge Neurons in Pretrained Transformers](#).

In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8493–8502, Dublin, Ireland. Association for Computational Linguistics.

<https://doi.org/10.18653/v1/2022.acl-long.581>

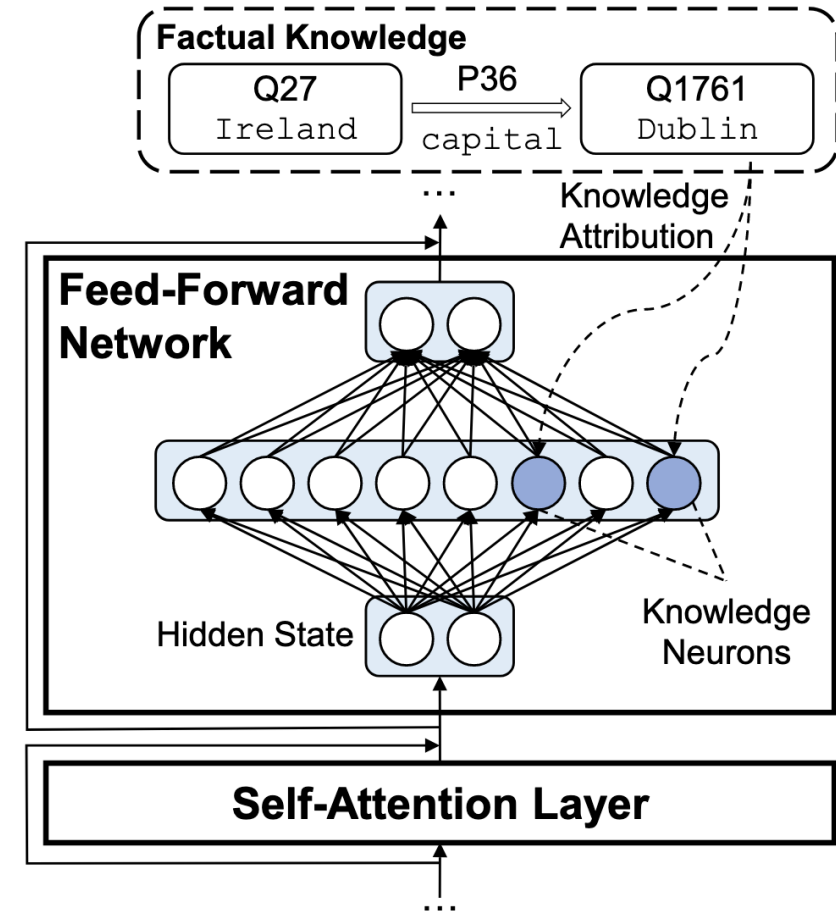


Figure 1: Through knowledge attribution, we identify knowledge neurons that express a relational fact.

Knowledge Neuron

$$\text{FFN}(H) = \text{gelu}(HW_1)W_2$$

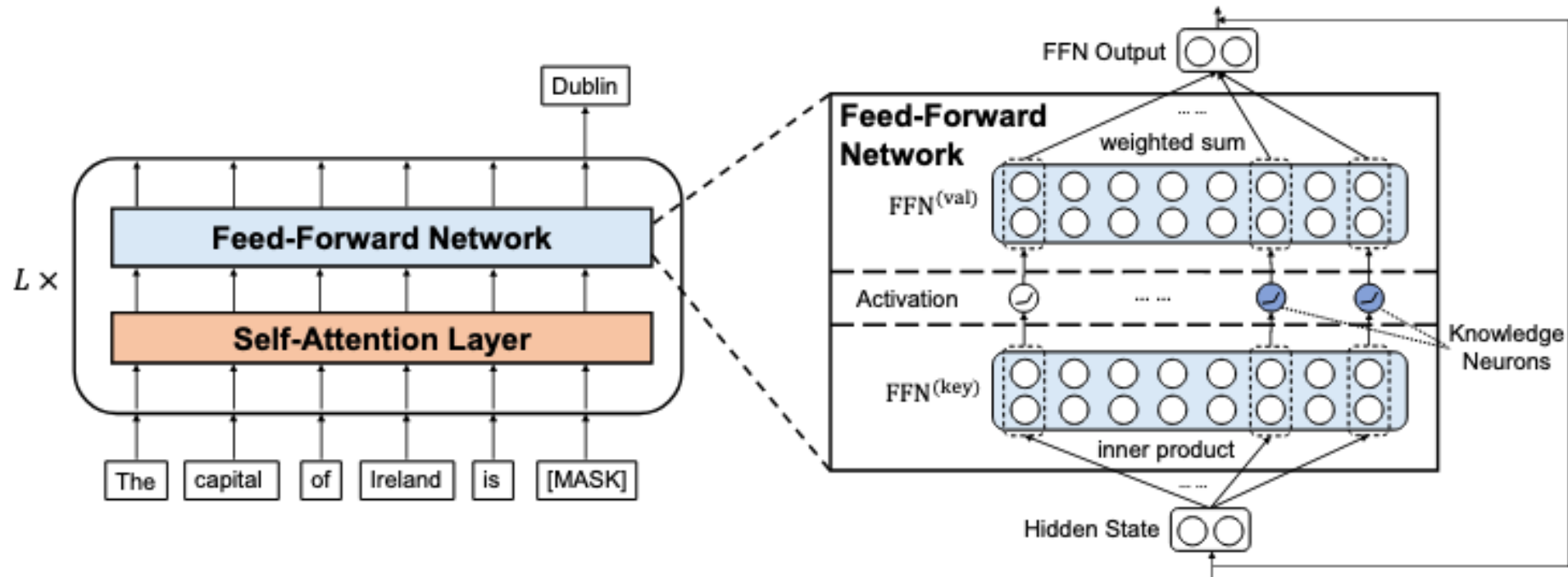


Figure 2: Illustration of how an FFN module in a Transformer block works as a key-value memory. The first linear layer $\text{FFN}^{(\text{key})}$ computes intermediate neurons through inner product. Taking the activation of these neurons as weights, the second linear layer $\text{FFN}^{(\text{val})}$ integrates value vectors through weighted sum. We hypothesize that knowledge neurons **in the FFN module** are responsible for expressing factual knowledge.

Knowledge Neuron – gradient-based attribution

Given the fact $\langle \text{Ireland, capital, Dublin} \rangle$, a possible query is “The capital of Ireland is ___”.

Given an input prompt x , we first define the model output $P_x(\hat{w}_i^{(l)})$ as the probability of the correct answer predicted by a pretrained model:

$$P_x(\hat{w}_i^{(l)}) = p(y^* | x, w_i^{(l)} = \hat{w}_i^{(l)}), \quad (4)$$

where y^* denotes the correct answer; $w_i^{(l)}$ denotes the i -th intermediate neuron in the l -th FFN; $\hat{w}_i^{(l)}$ is a given constant that $w_i^{(l)}$ is assigned to.

In order to calculate the attribution score of a neuron $\text{Attr}(w_i^{(l)})$, we gradually change $w_i^{(l)}$ from 0 to its original value $\bar{w}_i^{(l)}$ calculated by the pretrained model, and meanwhile integrate the gradients:

$$\text{Attr}(w_i^{(l)}) = \bar{w}_i^{(l)} \int_{\alpha=0}^1 \frac{\partial P_x(\alpha \bar{w}_i^{(l)})}{\partial w_i^{(l)}} d\alpha, \quad (5)$$

where $\frac{\partial P_x(\alpha \bar{w}_i^{(l)})}{\partial w_i^{(l)}}$ calculates the gradient of the model output with regard to $w_i^{(l)}$. Intuitively, as

4.3 Attribution Baseline

Our **baseline** method **takes the neuron activation value as the attribution score**, i.e., $\text{Attr}_{\text{base}}(w_i^{(l)}) = \bar{w}_i^{(l)}$, which measures how sensitive a neuron is to the input. After computing attribution scores,

Given a relational fact:

- (1) produce n diverse prompts;
- (2) for each prompt, calculate the knowledge attribution scores of neurons;
- (3) for each prompt, retain the neurons with attribution scores greater than the attribution threshold t , obtaining the coarse set of knowledge neurons;
- (4) considering all the coarse sets together, retain the knowledge neurons shared by more than $p\%$ prompts.

For example, relationship P176 (manufacturer) has templates:

- [X] is produced by [Y]
- [X] is a product of [Y]
- [Y] and its product [X]
- etc

Relational Facts	Neurons
$\langle \text{Ireland, capital, Dublin} \rangle$	$w_{2141}^{(9)}, w_{1122}^{(10)}$

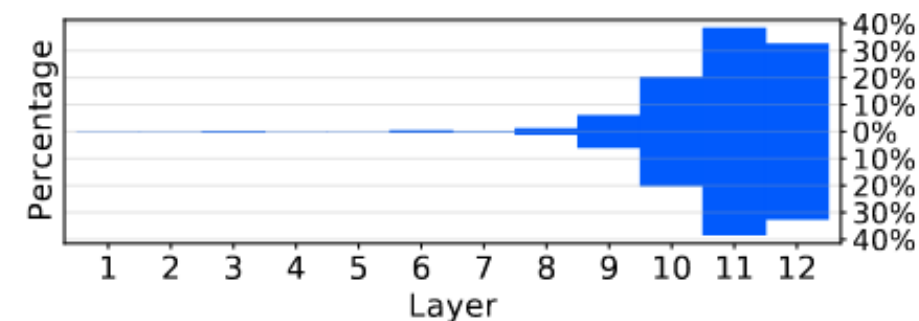


Figure 3: Percentage of knowledge neurons identified by our method in each Transformer layer.

Performed on BERT-base-cased

12 Transformer blocks, hidden size 768, FFN inner layer size 3072

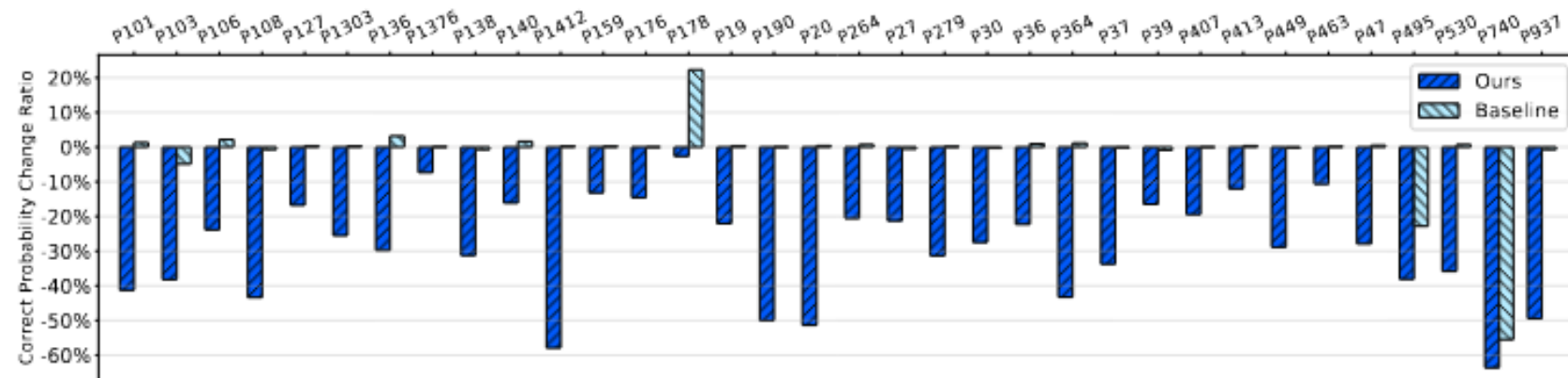


Figure 4: Results of suppressing knowledge neurons for various relations. Suppressing knowledge neurons decreases the correct probability by 29.03% on average. For the baseline, the decreasing ratio is 1.47% on average.

SHAP

- The goal of SHAP is to explain the prediction of an instance x by computing the contribution of each feature to the prediction.
- **Cooperative game theory:** The Shapley value, coined by Shapley (1953), is a method for assigning payouts to players depending on their contribution to the total payout. Players cooperate in a coalition and receive a certain profit from this cooperation.

Shapley value definition

The Shapley value is defined via a value function $val()$ of players in S .

The Shapley value of a feature value is its contribution to the payout, weighted and summed over all possible feature value combinations:

$$\phi_j(val) = \sum_{S \subseteq \{1, \dots, p\} \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} (val(S \cup \{j\}) - val(S))$$

Intuitive understanding of Shapley values

The feature values enter a room in random order. All feature values in the room participate in the game (= contribute to the prediction). The Shapley value of a feature value is the average change in the prediction that the coalition already in the room receives when the feature value joins them.

- SHAP is post-hoc and model-agnostic – and can be applied to various models, inc. Transformers (see TransSHAP).
- SHAP has a **solid theoretical foundation** in game theory. The prediction is **fairly distributed** among the feature values.
- We get **contrastive explanations** that compare the prediction with the average prediction.
- The fast computation makes it possible to compute the many Shapley values needed for the **global model interpretations**.

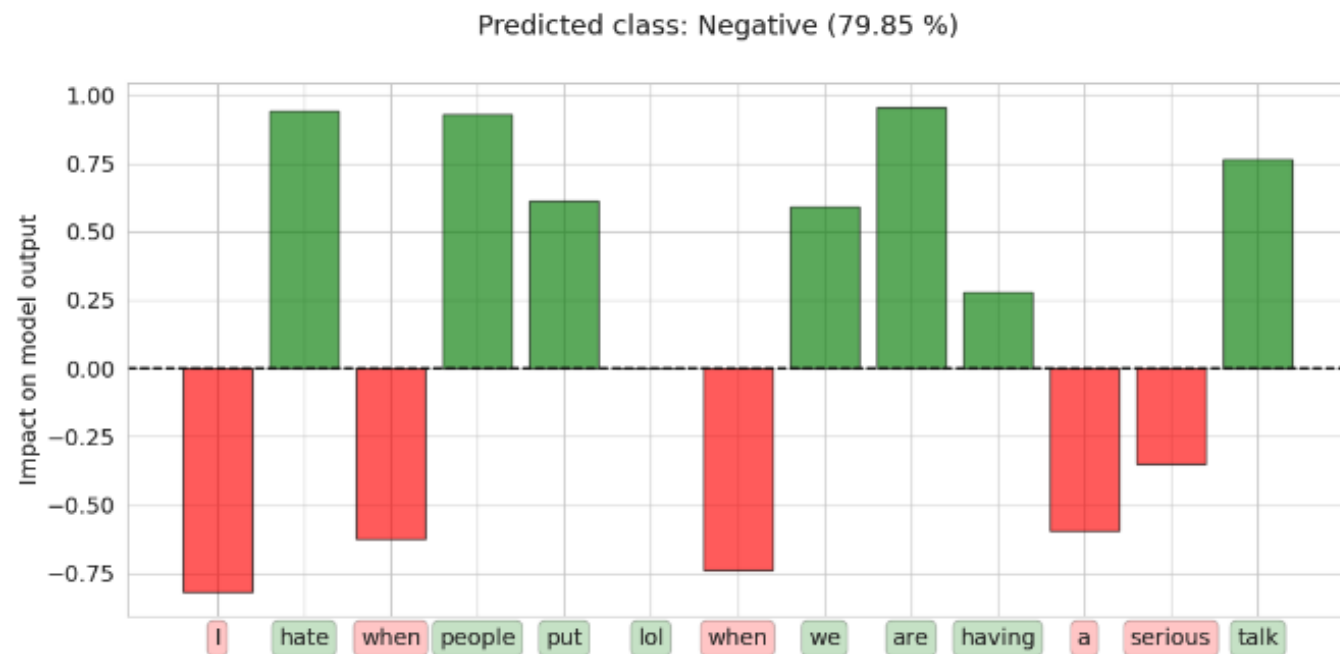


Figure 4: TransSHAP visualization of prediction explanations for negative sentiment. We obtained the features' contribution values with the SHAP method. It is evident that the word 'hate' strongly contributed to the negative sentiment classification, while the word 'lol' (laughing out loud) slightly opposed it.

Issues of Shapley values

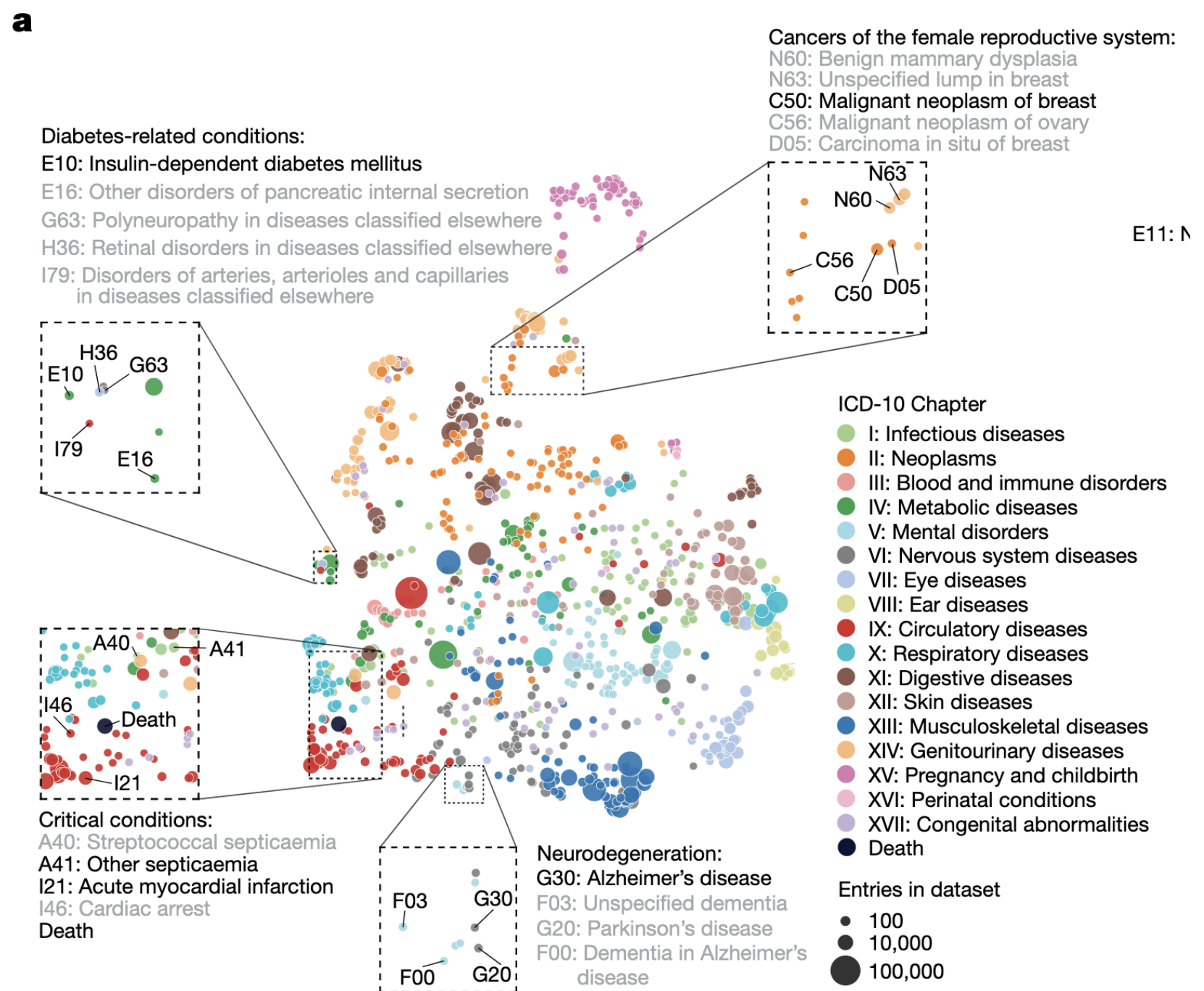
- The Shapley value **can be misinterpreted**. The Shapley value of a feature value is **not** the difference of the predicted value after removing the feature from the model training.
- Shapley value method suffers from **inclusion of unrealistic data instances** when features are correlated.
- It's **possible to create intentionally misleading interpretations** with SHAP, which can hide biases. Always be careful with **reference dataset** to create Shapley value.
- The Shapley value requires **a lot of computing time**, while more efficient adaptations like TreeShap were applied.

UMAP

- Intuition: a manifold learning approach
- Advantage: capturing the similarity within data; qualitative distance can be meaningful, e.g. relative distance between dense regions and clusters.
- Disadvantage: longer distance cannot be meaningful. The qualitative interpretation of the distance should always be verified.

UMAP

Figure from Shmatko, A., Jung, A. W., Gaurav, K., Brunak, S., Mortensen, L. H., Birney, E., Fitzgerald, T., & Gerstung, M. (2025). Learning the natural history of human disease with generative transformers. *Nature*, 1–9. <https://doi.org/10.1038/s41586-025-09529-3>



Explainability for LLMs

Global explanations

- Probing-based explanations
 - It is relevant to Week 4 (probing with knowledge facts) and Week 5 (Sparse Autoencoder).
- Neuron activation explanation
- Concept-based explanation
- Mechanistic explainability
 - We have read a paper together on Week 5 on Sparse Autoencoder based interpretability.

Probing-based explanations

- The probing technique refers to methods used to understand the knowledge that LLMs have captured.
 - Classifier-based probing:
 - The basic idea behind classifier-based probing is to train a shallow classifier on top of the pre-trained or fine-tuned language models such as BERT.
 - An example is the sparse autoencoder (<https://transformer-circuits.pub/2023/monosemantic-features>). We have discussed some related works in Week 5.
 - Parameter-Free probing:
 - Design datasets tailored to probe the knowledge capability of an LLM.
 - An example is the LAMA probe (<https://github.com/facebookresearch/LAMA>) We have discussed some related works in Week 4 on Language, Data and Knowledge.

Probing: metric learning to construct syntax tree

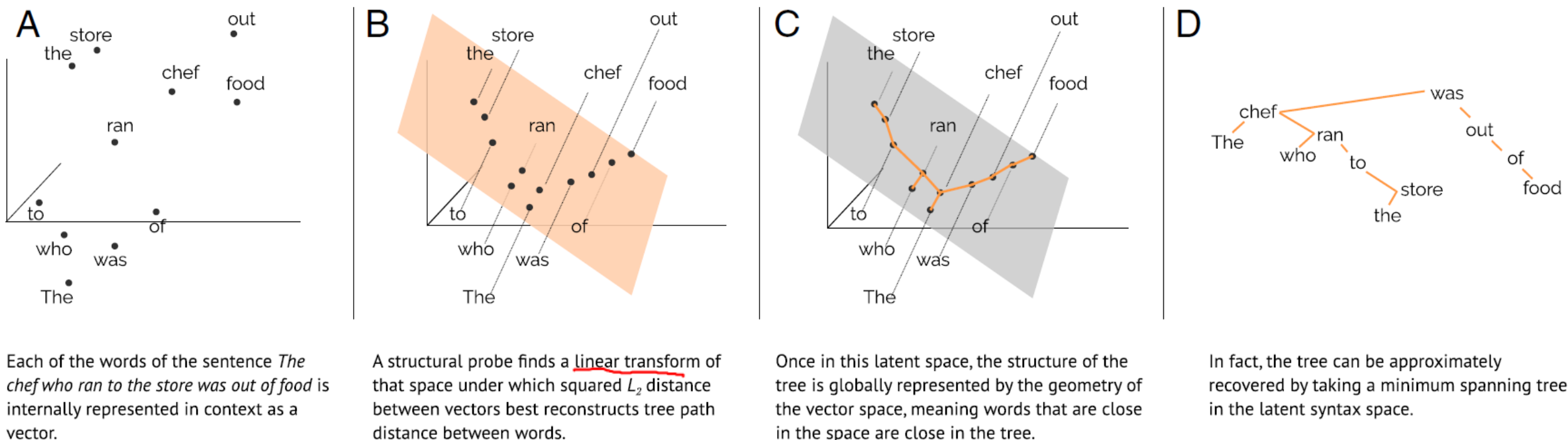


Fig. 7. (A–D) An overview of the structural probe method.

Probing: metric learning to construct syntax tree

sentence meaning. We note that L_2 distance on $\mathbb{R}^d$ can be parameterized with a positive semidefinite¹ matrix $A \in \mathbb{S}_+^{d \times d}$. All such matrices can in turn be represented as $A = B^\top B$ for some matrix $B \in \mathbb{R}^{k \times d}$, leading to a distance of the following form:

$$\begin{aligned} d_A(\mathbf{h}_i, \mathbf{h}_j)^2 &= (\mathbf{h}_i - \mathbf{h}_j)^\top A (\mathbf{h}_i - \mathbf{h}_j) \\ &= (B (\mathbf{h}_i - \mathbf{h}_j))^\top (B (\mathbf{h}_i - \mathbf{h}_j)) \\ &= \|B(\mathbf{h}_i - \mathbf{h}_j)\|_2^2. \end{aligned} \quad [4]$$

$$\arg \min_B \sum_{\ell} \frac{1}{|s^\ell|^2} \sum_{i,j} \left| d_{T^\ell}(w_i^\ell, w_j^\ell) - \|B(\mathbf{h}_i^\ell, \mathbf{h}_j^\ell)\|_2^2 \right|. \quad [5]$$

We approximate this objective through gradient descent to find B .

$$\arg \min_B \sum_{\ell} \frac{1}{|s^\ell|} \sum_i \left| \|w_i^\ell\|_{\text{depth}} - \|B\mathbf{h}_i^\ell\|_2^2 \right|. \quad [6]$$

Table 3. Results of structural probes on the WSJ SD test set (baselines in the top half, models hypothesized to encode syntax below)

Method	Distance		Depth	
	UUAS (%)	DSpr.	Root%	NSpr.
Linear	48.9	0.58	2.9	0.27
Proj0	59.8	0.73	64.4	0.75
BERT	81.7	0.87	90.1	0.89

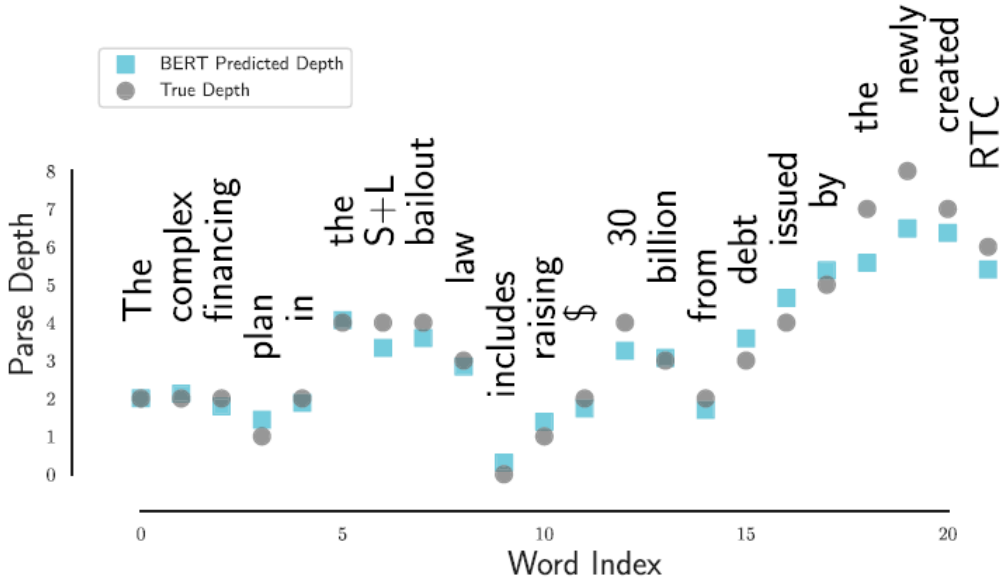
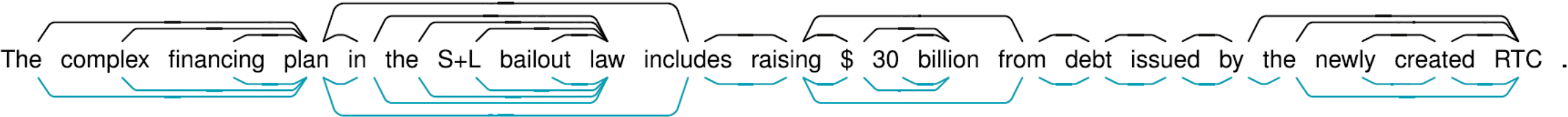


Fig. 10. A sentence with human-annotated parse depths (gray circles) and BERT-predicted parse depths (blue squares). Reprinted with permission from ref. 42, which is licensed under [CC BY 4.0](#).

Blue, below: structural probe tree on BERT; Black, above: Human-Annotated tree



Purple, below: structural probe tree on random control representation Proj0; Black, above: Human-Annotated tree

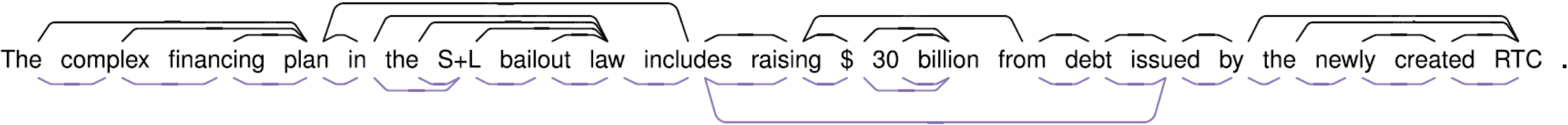


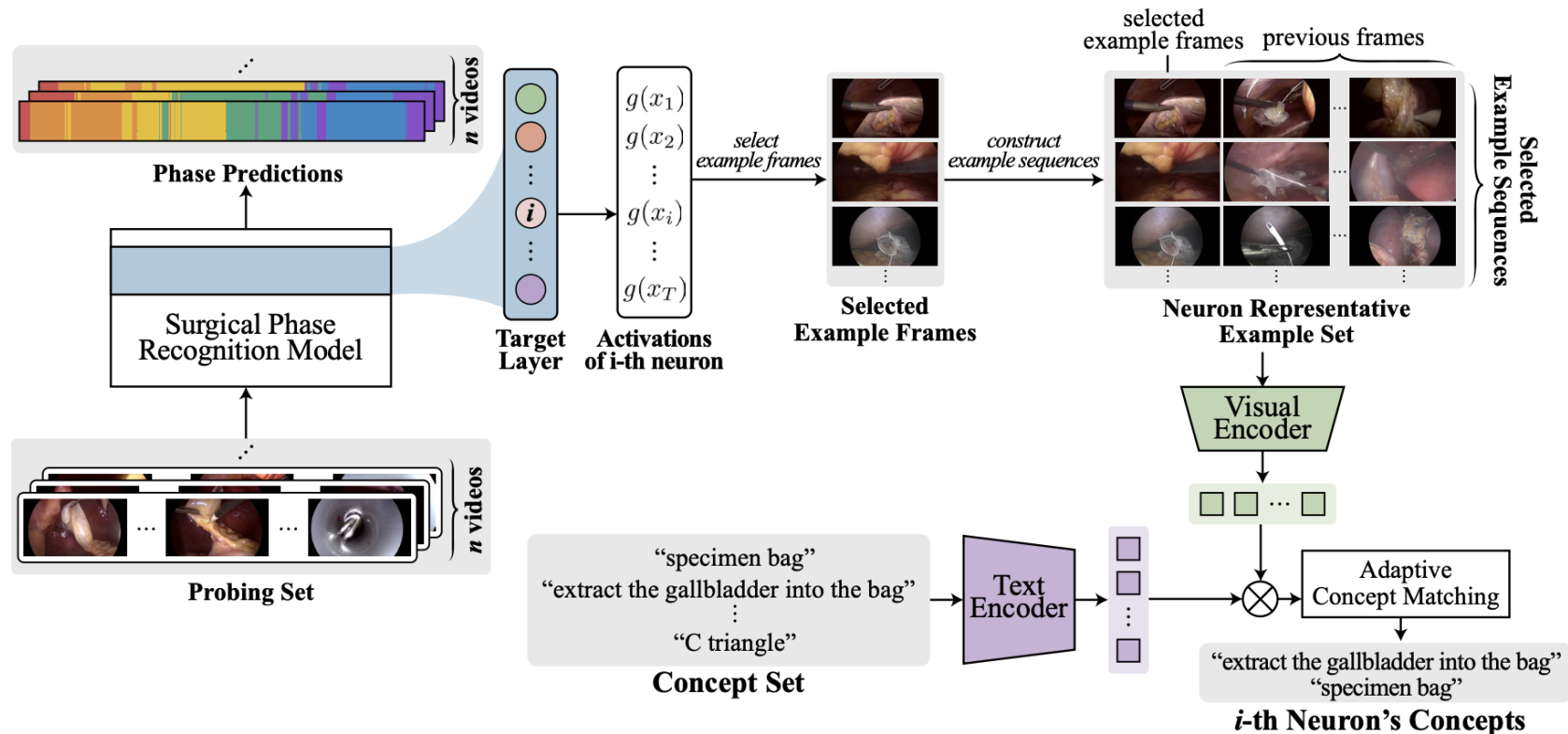
Fig. 8. Minimum-spanning trees resultant from structural probes on BERT and a random control representation Proj0 compared to the human-annotated parse tree. In the text sentence, “S+L” refers to American savings and loans banks and “RTC” refers to the Resolution Trust Corporation. Reprinted with permission from ref. 42, which is licensed under [CC BY 4.0](#).

Concept-based explanation

- Concept-based interpretability algorithms map the inputs to a set of concepts and measure importance score of each pre-defined concept to model predictions (Zhao et al., 2024).
- It is possible to extend concept-based to knowledge graph based explanation (Tiddi & Schlobach, 2022).

Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. Explainability for Large Language Models: A Survey. *ACM Trans Intell Syst Technol.* 2024 Feb 22;15(2):20:1-20:38.

Tiddi, I., & Schlobach, S. (2022). Knowledge graphs as tools for explainable machine learning: A survey. *Artificial Intelligence*, 302, 103627. <https://doi.org/10.1016/j.artint.2021.103627>



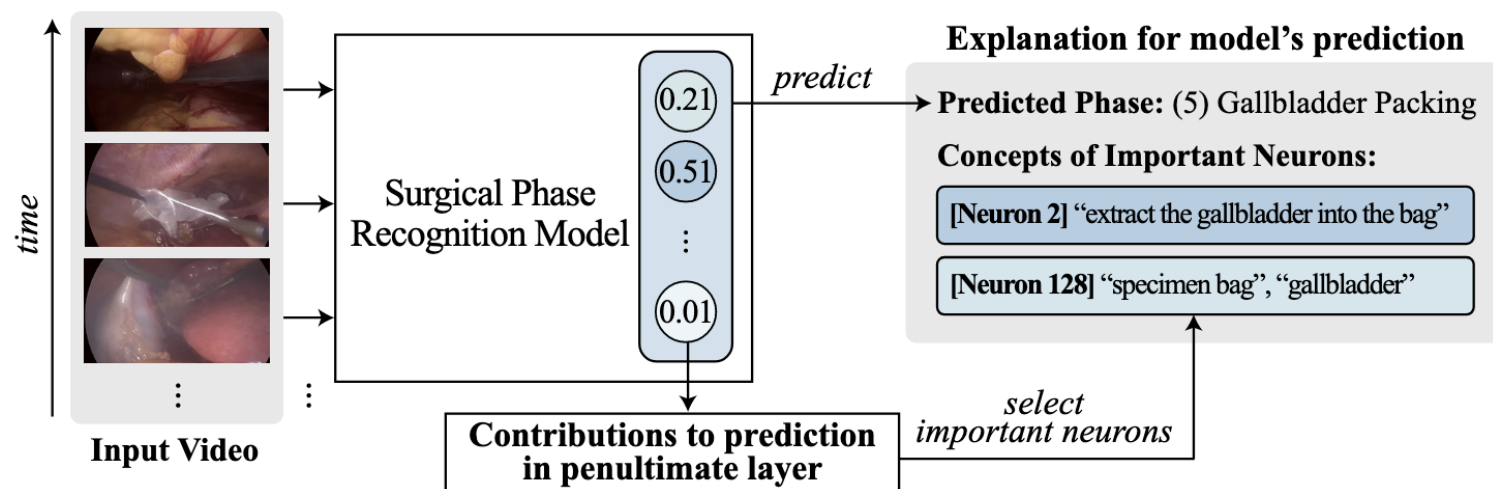
Project page: <https://ailab-kyunghee.github.io/SurgX/>

[1] K. Y. Kim, H. B. Kim, and S. T. Kim, 'SurgX: Neuron-Concept Association for Explainable Surgical Phase Recognition', in MICCAI 2025.

<https://papers.miccai.org/miccai-2025/0887-Paper2547.html>

[2] Y. H. Ahn, H. B. Kim, and S. T. Kim, 'WWW: A Unified Framework for Explaining What, Where and Why of Neural Networks by Interpretation of Neuron Concepts', Apr. 11, 2024, arXiv:2402.18956. doi: [10.48550/arXiv.2402.18956](https://doi.org/10.48550/arXiv.2402.18956).

Fig. 1. Neuron concept annotation model. To interpret the neuron representative examples are selected, and a text-based concept examples are constructed based on the neuron in the video probing set. Both the visual and text examples are projected into the vision-language model embedding space and identified through a similarity-based concept matching process.



Evaluation of Explainability

- Rationales: [data / text] snippets that support outputs.
- Plausibility: how well the rationales provided by models align with human rationales.
 - Sufficiency
 - Comprehensiveness
- Faithfulness: the degree to which provided rationales influenced the corresponding predictions.
 - Sufficiency
 - Comprehensiveness

Movie Reviews

In this movie, ... Plots to take over the world. The acting is great! The soundtrack is run-of-the-mill, but the action more than makes up for it

(a) Positive (b) Negative

e-SNLI

H A man in an orange vest leans over a pickup truck
P A man is touching a truck

(a) Entailment (b) Contradiction (c) Neutral

Commonsense Explanations (CoS-E)

Where do you find the most amount of leafs?

(a) Compost pile (b) Flowers (c) Forest (d) Field (e) Ground

Evidence Inference

Article Patients for this trial were recruited ... Compared with 0.9% saline, 120 mg of inhaled nebulized furosemide had no effect on breathlessness during exercise.

Prompt With respect to *breathlessness*, what is the reported difference between patients receiving *placebo* and those receiving *furosemide*?

(a) Sig. decreased (b) No sig. difference (c) Sig. increased

Figure 1: Examples of instances, labels, and rationales illustrative of four (out of seven) datasets included in ERASER. The ‘erased’ snippets are rationales.

Faithfulness:

- Comprehensiveness - were all features needed to make a prediction selected?
- Sufficiency - do the extracted rationales contain enough signal to come to a disposition?

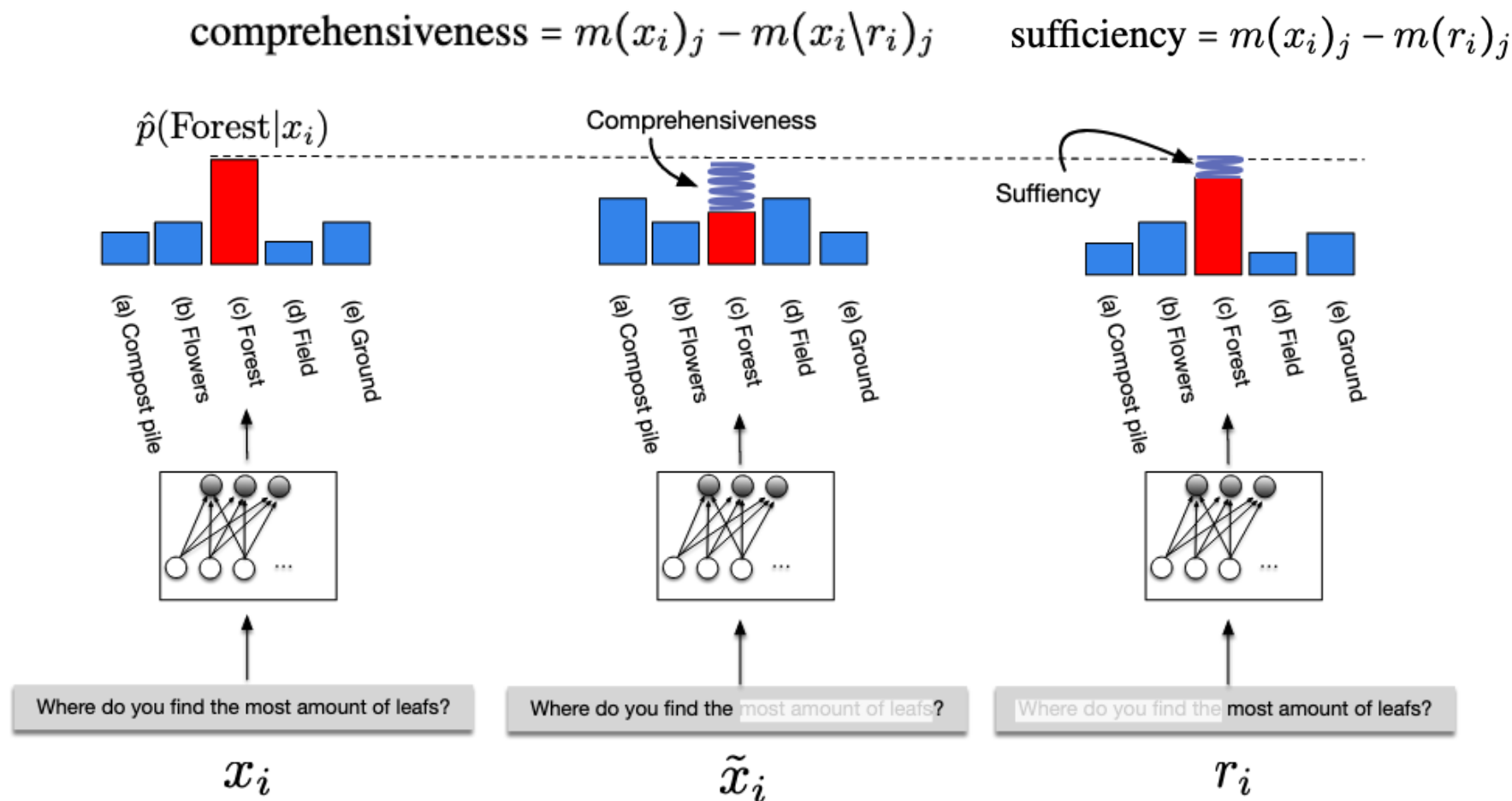


Figure 2: Illustration of faithfulness scoring metrics, *comprehensiveness* and *sufficiency*, on the Commonsense Explanations (CoS-E) dataset. For the former, erasing the tokens comprising the provided rationale ($\tilde{x}_i$) ought to decrease model confidence in the output ‘Forest’. For the latter, the model should be able to come to a similar disposition regarding ‘Forest’ using *only* the rationales r_i .

Model card

- meta-llama. Llama4 Model Card. GitHub.
https://github.com/meta-llama/llama-models/blob/main/models/llama4/MODEL_CARD.md. Apr 5, 2025.
- FacebookAI/roberta-base. Model Card.
<https://huggingface.co/FacebookAI/roberta-base>

Figure in M. Mitchell *et al.*, ‘Model Cards for Model Reporting’, in *Proceedings of the Conference on Fairness, Accountability, and Transparency*, Atlanta GA USA: ACM, Jan. 2019, pp. 220–229. doi: [10.1145/3287560.3287596](https://doi.org/10.1145/3287560.3287596).

Model Card
• Model Details. Basic information about the model. – Person or organization developing model – Model date – Model version – Model type – Information about training algorithms, parameters, fairness constraints or other applied approaches, and features – Paper or other resource for more information – Citation details – License – Where to send questions or comments about the model • Intended Use. Use cases that were envisioned during development. – Primary intended uses – Primary intended users – Out-of-scope use cases • Factors. Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others listed in Section 4.3. – Relevant factors – Evaluation factors • Metrics. Metrics should be chosen to reflect potential real-world impacts of the model. – Model performance measures – Decision thresholds – Variation approaches • Evaluation Data. Details on the dataset(s) used for the quantitative analyses in the card. – Datasets – Motivation – Preprocessing • Training Data. May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets. • Quantitative Analyses – Unitary results – Intersectional results • Ethical Considerations • Caveats and Recommendations

Figure 1: Summary of model card sections and suggested prompts for each.

Pre-class targeted reading task (for 10th March)

Investigate one of the techniques or directions as assigned, read something about it:

1. Attention-based explainability (Sect. 3.1.2)
 - <https://aclanthology.org/P19-3007/>
2. Concept-based explainability (Sect. 3.2.3)
 - <https://aclanthology.org/2022.acl-long.581/> (& for Neuron Activation)
 - <https://doi.ieeecomputersociety.org/10.1109/CVPR52733.2024.01043>
3. ERASER, Faithfulness evaluation of explainability
 - <https://aclanthology.org/2020.acl-main.408/>
4. SHAP (Surrogate Models in Sect. 3.1.1)
 - <https://christophm.github.io/interpretable-ml-book/shap.html>
 - <https://aclanthology.org/2021.hackashop-1.3.pdf>
5. UMAP
 - <https://arxiv.org/abs/1802.03426>
 - <https://pair-code.github.io/understanding-umap/>
 - <https://doi.org/10.1038/s43586-024-00363-x>
6. Other directions: Probing-based (Sect. 3.2.1), Neuron activation (Sect. 3.2.2), mechanistic interpretability (Sect. 3.2.4), **Natural Language Explanation** (Sect. 3.1.4 and 3.2.2), etc.

Think about five aspects:

- Intuition of the method
- A use case
- An advantage
- A disadvantage
- Feasibility to implement

Section numbers are from the survey:
Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. Explainability for Large Language Models: A Survey. ACM Trans Intell Syst Technol. 2024 Feb 22;15(2):20:1-20:38.
<https://doi.org/10.1145/3639372>

Initial Discussion

1. Re-introduce your project.
2. Is explainability needed in your proposed project? If so, what explainability approach would be suitable for your project?
3. What type of explainability you would need for your project?
4. How to evaluate explainability?

Workshop 11 March 2026 – explainability for LLMs

- See the Week 9 Question Sheet on ELE

Workshop

- See Week 9 Question Sheet on Explainability for LLMs below



[Week 9 Workshop Question Sheet - Explainability for LLMs](#)



PDF

Week 9 Question Sheet: Explainability for LLMs

1. Which explainability approach is used in AlphaFold [1] and Delphi-2M [2]? How do they suit?

[1] <https://doi.org/10.1038/s41586-021-03828-1>

[2] <https://doi.org/10.1038/s41586-025-09529-3>

2. Is explainability needed in your proposed project? If so, what explainability approach would be suitable for your project?

3. Design to implement an initial version of the explainability.

- For example, testing the explainability in Large Language Models using one of the approaches below
 - Attention visualisation, e.g. BERT-viz <https://github.com/jessevig/bertviz>
 - SHAP (see notebook on ELE), e.g. for transformers https://shap.readthedocs.io/en/latest/example_notebooks/text_examples/question_answering/Explaining%20a%20Question%20Answering%20Transformers%20Model.html
 - UMAP, e.g. https://github.com/gerstung-lab/Delphi/blob/main/evaluate_delphi.ipynb in [2]

4. Design an evaluation strategy of the explainability.

What **metrics** will be used? Does it need **annotated data**? Does it need **human evaluation**?

Discussion - Friday 13 March 2026

- Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017), 625–630. <https://doi.org/10.1038/s41586-024-07421-0>

Questions:

- What did the authors try to accomplish?
- What are the key elements of the approach?
- What can you use yourself?
- What other references do you want to follow?